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Suhyeon Lee

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PROFESSIONAL SUMMARY

Blockchain security researcher and engineer working on the security and economic design of decentralized systems. My research focuses on how adversarial behavior and incentive structures affect blockchain protocols, Layer 2 systems, PoW/PoS consensus, and smart contract mechanisms. I currently work on stablecoin infrastructure at ShardLab while conducting blockchain research as a Visiting Research Fellow at Hashed Open Research.

EDUCATION

Sep. 2017 – Feb. 2023 **Ph.D. in Cybersecurity**
Department of Information Security, School of Cybersecurity, Korea University
Supervisor: Prof. Seungjoo Kim
Thesis: Revisiting Conventional Wisdom in PoW and PoS-based Blockchains

Mar. 2012 – Feb. 2016 **B.Eng. in Cyber Defense**
Department of Cyber Defense, Korea University
Supervisor: Prof. Jong In Lim

PROFESSIONAL EXPERIENCE

July. 2026 – **Resarch Engineer**
ShardLab
- Building stablecoin infrastructure for Korea

Nov. 2025 – **Visiting Research Fellow**
Hashed Open Research
- Contributing to project-based research and reports on blockchain and cryptocurrency

June. 2023 – Mar. 2026 **Security Researcher**
Tokamak Network
- Security research focusing on Layer 2 blockchain systems

Sep. 2024 – Aug. 2025 **Adjunct Professor**
School of Cyber Security, Korea University
- Teaching a regular lecture SCS606 "Blockchain and Cryptocurrency"

Jul. 2019 – May. 2023 **Research Officer / Army Captain**
ROK Cyber Operations Command
- Participated in Korea-US joint military exercises and inter-agency cooperation.
- Led multiple cyber defense R&D projects including system design, cryptography, and AI technologies.

Jul. 2016 – Jul. 2019 **Researcher / Army Lieutenant**
2nd R&D Institute, Agency for Defense Development (ADD)
- Participated in R&D for collecting and processing Cyber Threat Intelligence (CTI).
- Participated in R&D for development of secure Real Time Operation System (RTOS) for weapon systems with high security verification (EAL≥4).

PROFESSIONAL SERVICE

Program Committee

WTSC 2026 – 10th Workshop on Trusted Smart Contracts (in association with Financial Cryptography 2026)
ICBC 2026 – IEEE International Conference on Blockchain and Cryptocurrency
ICDCS 2026 – 46th IEEE International Conference on Distributed Computing Systems

External Reviewer

Middleware 2026 – ACM/IFIP International Middleware Conference

Industry Presentations

ETHCC 2026, ETHCC 2025, ETHSofia 2024, ETHTokyo 2024, ETHVenice 2024

WORKING PAPERS (Preprint)

S. Lee et al., "Commit-Reveal²: A Deployable Randomness Beacon with a Robust Accountability Framework for Smart Contracts," *IEEE Transactions on Dependable and Secure Computing (TDSC)*, under revision.

S. Lee, and Y. Park, "Looking for Attention: Attention Test Design for Validator Monitoring in Optimistic Rollups," *arXiv:2505.24393*, June 2025, under review.

PUBLICATIONS – INTERNATIONAL CONFERENCES

S. Lee, and H. Kim, "Inside Qubic's Selfish Mining Attack in Monero: Evidence, Tactics, and Limits," *Advances in Financial Technologies (AFT 27)*, October 2026.

S. Lee, D. Nguyen, and D. Lee, "(Im)possibility of Incentive Design for Challenge-based Blockchain Protocols," *In Proceedings of the 10th Workshop on Trusted Smart Contracts (WTSC 2026), in association with Financial Cryptography 2026*, pp. TBD, St. Kitts, Saint Kitts and Nevis, March 6, 2026.

S. Lee, "180 Days After EIP-4844: Will Blob Sharing Solve Dilemma for Small Rollups?", *In Proceedings of the DICG Workshop, in association with the 45th IEEE International Conference on Distributed Computing Systems (ICDCS 2025)*, pp. TBD, Glasgow, UK, July 20–23, 2025.

S. Lee, and E. Gee, "Commit-Reveal²: Randomized Reveal Order Mitigates Last-Revealer Attacks in Commit-Reveal," *In Proceedings of the IEEE International Conference on Blockchain and Cryptocurrency (ICBC 2025)*, pp. TBD, Pisa, Italy, June 2-6, 2025.

S. Lee, "Hollow Victory: Analysis on Challenger's Incentive in Optimistic Rollup Dispute Game," *In Proceedings of the 9th Workshop on Trusted Smart Contracts (WTSC 2025), in association with Financial Cryptography 2025*, pp. TBD, Miyakojima, Japan, April 18, 2025.

H. Yang, **S. Lee**, and S. Kim, "A Tip for IOTA Privacy: IOTA Light Node Deanonymization via Tip Selection," *In Proceedings of the International Conference on IEEE Blockchain Conference (ICBC 2024)*, Dublin, Ireland, May 2024.

S. Lee, and S. Kim, "Do You Really Need to Disguise Normal Servers as Honeypots?," *In Proceedings of the 40th International Conference on IEEE Military Communications Conference (MILCOM 2022)*, pp. 166-172, Rockville, MD, US, November 2022.

S. Lee, D. Lee, and S. Kim, "Block Double-Submission Attack: Block Withholding Can Be Self-Destructive," *In Proceedings of the 4th International Conference on ACM Advances in Financial Technologies (AFT '22)*, Cambridge, MA, US, September 2022.

S. Lee and S. Kim, "Proof-of-Stake at Stake: Predatory, Destructive Attack on PoS Cryptocurrencies," *In Proceedings of ACM MobiCom 2020 Workshops – the 3rd International Workshop on Cryptocurrencies and Blockchains for Distributed Systems (CryBlock '20)*, pp. 7-11, Virtual Conference, September 2020.

S. Lee and S. Kim, "Blockchain for Cyber Defense: Will It Be as Good as You Think?" *the 28th DEF CON Hacking Conference (DEFCON 28) – the 2nd Blockchain Village*, Virtual Conference, August 2020.

S. Lee and S. Kim, "Countering Block Withholding Attack Efficiently," *In Proceedings of IEEE INFOCOM 2019 Workshops – the 2nd International Workshop on Cryptocurrencies and Blockchains for Distributed Systems (CryBlock '19)*, pp. 330-335, Paris, France, May 2019.

PUBLICATIONS – INTERNATIONAL JOURNALS

S. Lee, E. Jee, J. Lee, "Implementation of Cost-Effective Verification for Pietrzak's Verifiable Delay Function in Ethereum Smart Contracts," *Blockchain: Research and Applications* (I/F 5.6), In Press, June 2025.

S. Lee and S. Kim, "Shorting Attack: Predatory, Destructive Short Selling on Proof-of-Stake Cryptocurrencies," *Concurrency and Computation: Practice and Experience* (I/F 2), vol. 35, Issue 16, e6585, July 2023.

S. Lee and S. Kim, "Rethinking Selfish Mining under Pooled Mining," *ICT Express* (I/F 4.754), vol. 9, Issue 3, pp. 356-361, June 2023.

S. Lee, H. K. Kim, and K. Kim, "Ransomware Protection using the Moving Target Defense Perspective," *Computers & Electrical Engineering* (I/F 4.152), vol. 78. pp. 288-299, September 2019.

PATENTS

S. Kim and **S. Lee**, "Method for generating to tag for tracking document," submitted, May 2022.

H. Kim, K. Kim, and **S. Lee**, "Ransomware damage prevention method through extension randomization, recording medium and apparatus for performing the method," no. 10-2316404, October 2021.

S. Kim and **S. Lee**, "A block withholding attack detection device and its operation method," no. 10-2122824, June 2020.

S. Kim and **S. Lee**, "Mining pool management server capable of suppressing selfish mining strategies of competitive mining pools and method of operation," no. 10-2154354, September 2019.

D. Lee, B. Ko, C. Lim, D. Kim, H. Ryu, **S. Lee**, and J. Park, "Device and method for extracting web page information through inference and learning of html tag information," no. 10-2018-0137256, December 2018.

INVITED TALKS

Suhyeon Lee, "Research and Experience in Cyber Defense," Capstone Project in the Department of Cyber Defense, Korea University, Seoul, April 2023.

Suhyeon Lee, "Threat Modeling for Cybersecurity of Industrial Products," 78ResearchLab, Gangnam-gu, Seoul, October 2022.

Suhyeon Lee, "Shorting Attack on Proof-of-Stake Cryptocurrencies," Monthly Crypto Seminar in Ripple, Virtual, July 2022.

Suhyeon Lee, "Blockchain Technology for Military Cyber Security," The 28th Network Security Conference Korea (NetSec-KR), COEX, Seoul, Korea, April 2022.

Suhyeon Lee, "K-Cyber Security Challenge Winner's Talk," Conference Center of the Federation of Korean Industries, Seoul, December 2019.

LANGUAGE & ADDITIONAL SKILLS

- Language: Korean (native), English (proficient)
- Programming: C, Python (familiar)
- Certification: Certified Information Systems Security Professional (CISSP), Certification for English Mediated Instruction (Awarded by the Center for Teaching and Learning of Korea University)

AWARDS & HONORS

Dec. 2019 **1st Prize in Korea Security Data Challenge (AI Section)**
Granted by Korea Internet & Security Agency (KISA)

2012 – 2016 **Full-tuition Military Scholarship**
Granted by Ministry of National Defense, Republic of Korea

2012 – 2016 **Merit-based Scholarship**
Granted by Korea University, Republic of Korea

2017 – 2019 **General Scholarships**
Granted by Korea University, Republic of Korea

REFERENCES

Professor. Seungjoo Kim (PhD Supervisor)
- School of Cybersecurity, Korea University, Republic of Korea
- President of Office of Digital Information, Korea University
- Member of Republic of Korea Presidential Defense Innovation Committee

Professor. Huy Kang Kim
- School of Cybersecurity, Korea University, Republic of Korea
- Director of Hacking and Countermeasure Research Lab (https://ocslab.hksecurity.net)

Senior Researcher. Donghwan Lee
- Agency for Defense Development, Republic of Korea

Chair-Professor. Jongin Lim
- School of Cybersecurity, Korea University, Republic of Korea
- Advisor of Kim & Chang Law Firm, Republic of Korea
- Former Special Advisor to the President for National Security